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Interference fits –
Part 1: Calculation and design rules for cylindrical self-locking pressfits, English translation of DIN 7190-1:2017-02

Pressverbände –
Teil 1: Berechnungsgrundlagen und Gestaltungsregeln für zylindrische Pressverbände, Englische Übersetzung von DIN 7190-1:2017-02

Assemblages frettés –
Partie 1: Bases de calculs et règles de construction pour assemblages frettés cylindriques autoadhérents,
Traduction anglaise de DIN 7190-1:2017-02

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Foreword

This document has been prepared by Working Committee NA 060-34-32 AA "Shafts and shaft-hub connections" of Section "Gear" of *DIN-Normenausschuss Maschinenbau (NAM)* (DIN Standards Committee Mechanical Engineering).

This document contains the revised edition of DIN 7190:2001-02. The new standard, DIN 7190-2, expands the scope of this series of standards to include conical, self-locking interference fits. In this way, it provides the basis for calculations for this type of interference fits, which are becoming ever more important, especially in terms of assembly with larger shaft diameters.

The calculation equations in the previous edition, DIN 7190:2001-02, for the elastic-plastic design of interference fits have been retained. Only the smoothening factor has been halved, as experience and new research show that the previous procedure was too conservative.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. DIN [and/or DKE] shall not be held responsible for identifying any or all such patent rights.

Amendments

This standard differs from DIN 7190:2001-02 as follows:

a) the standard has been divided into two parts: DIN 7190-1 replaces DIN 7190:2001-02 in terms of content and DIN 7190-2 expands the scope of this series of standards to include conical, self-locking interference fits;

b) the document has been editorially revised;

c) the normative references have been updated;

d) the smoothening factor has been reduced from 0,8 to 0,4 to reflect practical experience.

Previous editions

DIN 7182-3: 1942-06, 1977-08
DIN 7190: 1943-08, 1977-08, 1981-03, 1988-07, 2001-02

1 Scope

This standard specifies the calculation bases for interference fits with cylindrical effective surfaces whose parts consist of metallic materials. This standard applies primarily to mechanical engineering. It can be applied by analogy in other fields (e.g. precision engineering).

The objective of this standard is to improve the reliability of the interference fits and reduce manufacturing costs.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

DIN 743-1, *Calculation of load capacity of shafts and axles — Part 1: General*

DIN 743-2, *Calculation of load capacity of shafts and axles — Part 2: Theoretical stress concentration factors and fatigue notch factors*

DIN 743-3, *Calculation of load capacity of shafts and axles — Part 3: Strength of materials*

DIN EN 10052, *Vocabulary of heat treatment terms for ferrous products*

DIN EN ISO 286-1, *Geometrical product specification (GPS) — ISO code system for tolerances on linear sizes — Part 1: Basis of tolerances, deviations and fits*

DIN EN ISO 1101, *Geometrical product specifications (GPS) — Geometrical tolerancing — Tolerances of form, orientation, location and run-out*

DIN EN ISO 1302, *Geometrical Product Specifications (GPS) — Indication of surface texture in technical product documentation*

DIN EN ISO 4287, *Geometrical Product Specifications (GPS) — Surface texture: Profile method — Terms, definitions and surface texture parameters*

DIN EN ISO 6892-1, *Metallic materials — Tensile testing — Part 1: Method of test at room temperature*

3 Symbols and abbreviated terms

3.1 Symbols

Symbol	Term	Unit
A	elongation after fracture	%
a	axial projection	mm
b	bore shoulder	mm
A_o	upper deviation of the joint diameter D_F	mm
A_u	lower deviation of the joint diameter D_F	mm
D_a	outer diameter	mm
D_i	inner diameter	mm
D_F	diameter of the joint (nominal dimension)	mm
D_P	plasticity diameter	mm
D_w	diameter of the shaft shoulder	mm
E	elasticity modulus	N/mm^2
e	natural logarithm base (Euler number)	—
F_{ax}	transferable axial force	N
F_e	press-in force	N
g_F	smoothening factor	—
K	auxiliary variable for the design of elastic interference fits	—
l_F	length of the joint	mm
l_e	length of the assembly chamfer	mm
n	rotational speed	1/s
p	joint pressure at rest	N/mm^2
p_n	joint pressure at rotational speed n	N/mm^2
p_{PA}	joint pressure limit for fully plastic outer part	N/mm^2
p_{PI}	joint pressure limit for fully plastic inner part	N/mm^2
Q	diameter ratio	—
q_A	ring zone of the outer part	mm^2
q_{PA}	plastic ring zone of the outer part	mm^2
Ra	arithmetical roughness average	µm
R_{eL}	lower yield strength	N/mm^2
$R_{p,0,2}$	0,2-% elongation limit	N/mm^2
Rz	mean height of roughness of the joint surface	µm

Symbol	Term	Unit
r	transition radius	mm
S_P	safety against plastic strain (desired value)	—
S_r	safety against sliding (desired value)	—
T	transferable torque	Nmm
u	peripheral speed of the outer contour of the outer part	mm/s
u_{ab}	peripheral speed of the outer contour of the outer part, where the outer part lifts up	mm/s
U	interference	mm
U_i	actual interference	mm
U_{max}	maximum interference that can be fitted	mm
U_g	maximum interference	mm
$U_{s,\vartheta}$	clearance	mm
U_k	minimum interference	mm
U_F	interference during joining	mm
U_w	effective interference	mm
Z	reduction in area after fracture	%
α	coefficient of linear expansion	1/°C
ζ	reference plasticity diameter	—
ζ_{zul}	permissible reference plasticity diameter of the outer part	—
ϑ_A	temperature of the outer part while joining	°C
ϑ_{Aerf}	required temperature of the outer part while joining	°C
ϑ_{Azul}	permissible temperature of the outer part while joining	°C
ϑ_I	temperature of the inner part while joining	°C
ϑ_R	room temperature	°C
μ	Poisson's ratio	—
ν	coefficient of adhesion (general)	—
ν_l	coefficient of adhesion for separation	—
ν_r	coefficient of adhesion for sliding	—
ξ_w	reference effective interference	—
ρ	density	kg/dm^3
φ	chamfer angle	degree

3.2 Indices

Index	Term
A	outer part
I	inner part
l	in longitudinal direction
u	in circumferential direction

4 Calculation of interference fits

4.1 Basic principles

The calculation methods specified in this standard are applicable to interference fits with the same constant axial length of the inner and outer parts (see Figure 1).

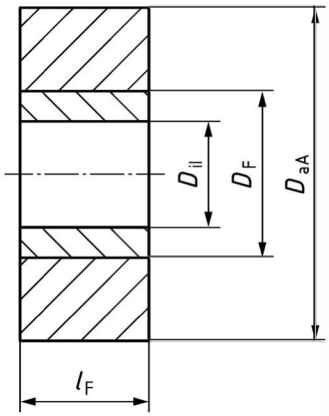


Figure 1 — Calculation model

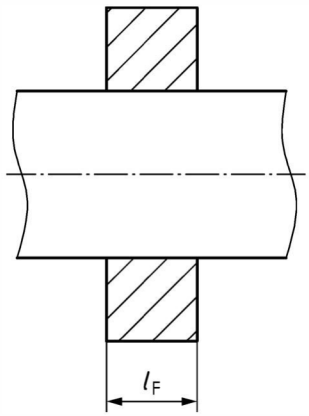


Figure 2 — Real interference fit

The calculation methods can also be used by approximation for interference fits as shown in Figure 2. However, the excessive stresses at the hub edge [1], [15] area are not covered.

With the joint pressure, p, at rest, the desired safety against sliding, S_r, and the coefficient of adhesion, ν_{ru}, (cf. Clause 5), the transferable torque is

$$T = \frac{\pi}{2} D_F^2 l_F \nu_{ru} \frac{p}{S_r} \tag{1}$$

For the transferable axial force, the following applies:

$$F_{ax} = \pi D_F l_F \nu_{rl} \frac{p}{S_r} \tag{2}$$

The diameter D_F of the joint surface is the nominal size of the parts of the interference fit to be joined. The actual interference is measured from the actual dimensions D_{iA} and D_{aI}:

$$U_i = |D_{iA} - D_{aI}| \tag{3}$$

Fits for the inner and the outer part are generally defined with the upper deviations A_{oA} and A_{oI}, and the lower deviations A_{uA} and A_{uI}. With the same nominal size for the shaft and hub, the following applies for the maximum interference:

$$U_g = A_{oI} - A_{uA} > 0 \tag{4}$$

and for the minimum interference:

$$U_k = A_{uI} - A_{oA} > 0 \tag{5}$$

The actual interference lies in the range

$$U_k \leq U_i \leq U_g \tag{6}$$

As a result of the smoothing of roughness peaks during joining, only the effective interference, U_W, remains available in the joined state. In the absence of empirical values, the following is applicable for force fits and shrink fits with the smoothening factor g_F = 0,4

$$U_w = U - 0{,}4(Rz_A + Rz_I) \tag{7}$$

Depending on the application, the minimum, maximum or actual interference shall be used for *U*.

If the numerical values for the arithmetical mean deviation of the profile *Ra* (formerly the arithmetical roughness average *Ra*) according to DIN EN ISO 4287 are specified, then the mean maximum height of the profile *Rz* (mean height of roughness *Rz*) determined according to Table 1 can be used here.

Table 1 — Comparison of arithmetical roughness average, *Ra*, with mean height of roughness, *Rz*; annotation according to DIN EN ISO 1302

Ra in µm		Ra 0,8	Ra 1,6	Ra 3,2
Rz in µm	from	3,15	6,3	12,5
	to	10	20	31,5
Mean height of roughness *Rz* in µm		Rz 6,3	Rz 12,5	Rz 20

Clause 8 contains information on the calculation of the joinable fit of shrink fits.

Depending on the magnitude of the stresses to which the outer and inner parts are subjected as a result of joining, a differentiation is made between purely elastic, elastic-plastic and fully plastic interference fits. In the case of purely elastic interference fits, plastic strain of the inner and the outer parts cannot occur. In elastic-plastic interference fits, there are regions in the inner and/or outer part that are subjected to elastic and plastic stresses. In fully plastic interference fits, unlimited plastic strain can occur in the inner and/or outer part. For this reason, interference fits subjected to fully plastic stresses are not permissible.

For the calculations given in this standard, the lower yield strength, R_{eL}, is used as the ultimate strength of the material. In reinforcing materials, R_{eL} shall be replaced by $R_{p0,2}$, provided the material is sufficiently ductile. To assess the ductility, the characteristic values elongation after fracture, *A*, and reduction in area after fracture, *Z*, determined according to DIN EN ISO 6892-1, can be used.

The guideline values for sufficient ductility are

$$A \geq 10\ \% \tag{8}$$

$$Z \geq 30\ \% \tag{9}$$

If these guideline values are not met, brittle material must be expected. In this case only a purely elastic design is permitted.

For the design of interference fits, the diameter ratios

$$Q_{\mathrm{A}} = \frac{D_{\mathrm{F}}}{D_{\mathrm{aA}}} \tag{10}$$

$$Q_{\mathrm{I}} = \frac{D_{\mathrm{iI}}}{D_{\mathrm{F}}} \tag{11}$$

and the reference effective interference

$$\xi_{\mathrm{w}} = \frac{U_{\mathrm{w}}}{D_{\mathrm{F}}} \tag{12}$$

are required.

While designing interference fits, there are two calculation methods:

Calculation method 1:

Joint pressure, p, is given; effective interference, U_{w}, and interference, U, are sought.

Calculation method 2:

Interference, U, and effective interference, U_{w}, are given; joint pressure, p, is sought.

In both methods of calculation, it is first checked whether the specified quantities (in calculation method 1: joint pressure, p; in calculation method 2: interference U, and effective interference U_{w}) result in permissible stresses in the inner and outer parts. Finally, the required determinants are calculated.

4.2 Calculation of interference fits subjected to purely elastic stresses

4.2.1 General

For the calculation, the auxiliary variable

$$K = \frac{E_{\mathrm{A}}}{E_{\mathrm{I}}}\left(\frac{1+Q_{\mathrm{I}}^2}{1-Q_{\mathrm{I}}^2} - \mu_{\mathrm{I}}\right) + \frac{1+Q_{\mathrm{A}}^2}{1-Q_{\mathrm{A}}^2} + \mu_{\mathrm{A}} \tag{13}$$

is introduced. In the case of a solid inner part ($Q_{\mathrm{I}} = 0$), the following applies:

$$K = \frac{E_{\mathrm{A}}}{E_{\mathrm{I}}}(1-\mu_{\mathrm{I}}) + \frac{1+Q_{\mathrm{A}}^2}{1-Q_{\mathrm{A}}^2} + \mu_{\mathrm{A}} \tag{14}$$

If $Q_{\mathrm{I}} = 0$, $E_{\mathrm{A}} = E_{\mathrm{I}}$ and $\mu_{\mathrm{A}} = \mu_{\mathrm{I}}$, then K need not be calculated (cf. Equations (19) and (25)).

4.2.2 Calculation method 1 for a specified joint pressure p

To ensure the desired safety against plastic strain, S_P, the following shall be applicable:

for the outer part:

$$p \leq \frac{1 - Q_A^2}{\sqrt{3}\, S_{PA}} R_{eLA} \tag{15}$$

for a hollow inner part ($Q_I > 0$):

$$p \leq \frac{1 - Q_I^2}{\sqrt{3}\, S_{PI}} R_{eLI} \tag{16}$$

for a solid inner part ($Q_I = 0$):

$$p \leq \frac{2\, R_{eLI}}{\sqrt{3} S_{PI}} \tag{17}$$

The reference effective interference, ξ_w, required for calculation of the joint pressure, p, is

$$\xi_w = K \frac{p}{E_A} \tag{18}$$

Together with Equations (7) and (12), this gives the permissible interference, U.

In the special case that the elasticity constants of a solid inner part are equal to those of the outer part ($E_I = E_A = E; \mu_A = \mu_I = \mu$), the calculation of the auxiliary value K according to Equation (13) is omitted and Equation (18) is replaced by

$$\xi_w = \frac{2}{1 - Q_A^2} \frac{p}{E} \tag{19}$$

4.2.3 Calculation method 2 for a specified interference U

From the specified interference, U, the effective interference, U_w, is calculated using Equation (7) and from this, the reference effective interference, ξ_w, is calculated using Equation (12).

For the outer part to meet the desired safety against plastic strain, S_P, the following shall be applicable

$$\xi_w \leq K \frac{1 - Q_A^2}{\sqrt{3}\, S_{PA}} \frac{R_{eLA}}{E_A} \tag{20}$$

and correspondingly for a hollow inner part

$$\xi_w \leq K \frac{1 - Q_I^2}{\sqrt{3}\, S_{PI}} \frac{R_{eLI}}{E_A} \tag{21}$$

If a solid inner part has the same elasticity constants as the outer part ($E_A = E_I = E; \mu_A = \mu_I = \mu$), then the following replaces Equation (20):

$$\xi_w \leq \frac{2\, R_{eLA}}{\sqrt{3}\, S_{PA} \cdot E} \tag{22}$$

and instead of Equation (21), the following applies, taking Equations (17) and (19) into account:

$$\xi_{\mathrm{w}} \leq \frac{4\,R_{\mathrm{eLI}}}{\sqrt{3}\,(1-Q_{\mathrm{A}}^2)\,S_{\mathrm{PI}}\cdot E} \tag{23}$$

For elastic stress in the outer and the inner part to be present, both the conditions (20) and (21) or (22) and (23) are required to be met.

The joint pressure, p, corresponding to the reference effective interference, ξ_{w}, is calculated from

$$p = \frac{\xi_{\mathrm{w}}\,E_{\mathrm{A}}}{K} \tag{24}$$

With a solid inner part and equal elasticity constants ($E_{\mathrm{I}} = E_{\mathrm{A}} = E; \mu_{\mathrm{A}} = \mu_{\mathrm{I}} = \mu$), it becomes

$$p = \frac{1-Q_{\mathrm{A}}^2}{2}\,E\,\xi_{\mathrm{w}} \tag{25}$$

4.3 Calculation of interference fits subjected to elastic-plastic stresses

4.3.1 General

This standard specifies a simple calculation method for interference fits subject to elastic-plastic stress which is restricted in its scope of application and for which the following prerequisites are required to be met to be applicable:

a) the inner part shall be solid ($Q_{\mathrm{I}} = 0$);

b) the inner and the outer parts shall have equal elasticity constants ($E_{\mathrm{I}} = E_{\mathrm{A}} = E; \mu_{\mathrm{A}} = \mu_{\mathrm{I}} = \mu$).

For interference fits for which the conditions a) and/or b) are not applicable, the design methods [2] and [3] are specified.

Elastic-plastic stresses cannot occur in a solid inner part, however they can occur in the outer part. A solid inner part is subjected to either pure elastic or fully plastic stresses.

4.3.2 Calculation method 1 for a specified joint pressure p

For the outer part to be subjected to elastic-plastic stresses and the desired safety against fully plastic stresses, S_{PA}, to be ensured, the joint pressure shall be in the range

$$\frac{1}{\sqrt{3}}(1-Q_{\mathrm{A}}^2)\,R_{\mathrm{eLA}} < p \leq \frac{p_{\mathrm{PA}}}{S_{\mathrm{PA}}} \tag{26}$$

The joint pressure limit, p_{PA}, for fully plastic stressing of the outer part results is given by the equations

$$p_{\mathrm{PA}} = 2\,R_{\mathrm{eLA}}/\sqrt{3} \text{ für } Q_{\mathrm{A}} < 1/e \approx 0{,}368 \tag{27}$$

$$p_{\mathrm{PA}} = -2\,R_{\mathrm{eLA}}\ln Q_{\mathrm{A}}/\sqrt{3} \text{ für } Q_{\mathrm{A}} \geq 0{,}368 \tag{28}$$

To prevent the inner part from being subjected to fully plastic stress, the following is required:

$$p \leq \frac{p_{\mathrm{PI}}}{S_{\mathrm{PI}}} \tag{29}$$

with the joint pressure limit for the fully plastic stress of the inner part, p_{PI},

$$p_{PI} = \frac{2}{\sqrt{3}} R_{eLI} \tag{30}$$

Often, the connecting parts provide a sufficient impediment to deformation of the inner part, so it is possible to estimate a lower desired safety against unlimited plastic deformation than for the outer part ($S_{PI} < S_{PA}$) [4].

If elastic-plastic stresses occur, an internal plastic zone is formed in the outer part, which is separated from the remaining external elastic zone by a cylindrical surface of plasticity diameter D_{PA} (see Figure 3).

The reference plasticity diameter

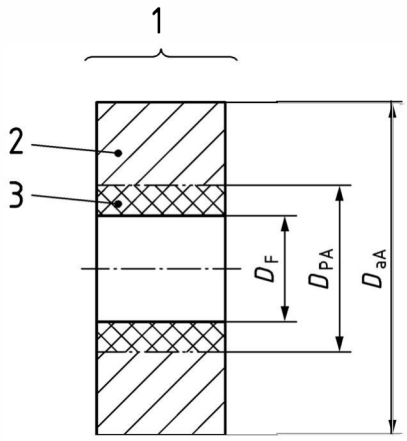
$$\zeta = \frac{D_{PA}}{D_F} \tag{31}$$

is determined by solving the transcendent equation

$$2 \ln \zeta - (Q_A \zeta)^2 + 1 - \sqrt{3} \frac{p}{R_{eLA}} = 0 \tag{32}$$

The reference plasticity diameter, ζ, shall satisfy the condition

$$1 \leq \zeta \leq \frac{1}{Q_A} \tag{33}$$



Key

1 outer part
2 elastic zone
3 plastic zone

Figure 3 — Hub subjected to elastic-plastic stresses

An approximate method for the calculation of ζ is specified in Annex B.

For the specified joint pressure, p, the required reference effective interference, ξ_w, (see Equation (12)) becomes

$$\xi_w = \frac{2}{\sqrt{3}} \zeta^2 \frac{R_{eLA}}{E} \tag{34}$$

Finally, the proportion of the plastic ring zone, q_{PA}, in relation to the overall cross section of the outer part, q_A, shall be checked.

$$\frac{q_{PA}}{q_A} = \frac{(\zeta^2 - 1)\, Q_A^2}{1 - Q_A^2} \tag{35}$$

For heavy duty interference fits in mechanical engineering, the following empirical value should not be exceeded:

$$\frac{q_{PA}}{q_A} = 0{,}30$$

4.3.3 Calculation method 2 for a specified interference, U

From the specified interference U, the effective interference U_w is calculated according to Equation (7), and from this, the reference effective interference, ξ_w, is calculated according to Equation (12). If

$$(1 - Q_A^2)\frac{R_{eLA}}{2} \geq R_{eLI} \tag{36}$$

is true, then an elastic-plastic design of the interference fit is not possible, as the inner part becomes fully plastic if interference is present while the outer part remains subjected to purely elastic stresses.

For purposes of further calculation, a distinction shall be made between two different cases:

Case 1:

If the following is valid

$$\frac{p_{PI}}{S_{PI}} \leq \frac{p_{PA}}{S_{PA}} \tag{37}$$

with the joint pressure limit, p_{PI}, of the fully plastic stress in the inner part according to Equation (30) and the joint pressure limit, p_{PA}, of the fully plastic stress in the outer part according to Equation (28), then the permissible reference plasticity diameter, ζ_{zul}, of the outer part is determined by solving the transcendent Equation (32) using

$$p = \frac{p_{PI}}{S_{PI}} \tag{38}$$

Case 2:

If the following is valid

$$\frac{p_{PA}}{S_{PA}} < \frac{p_{PI}}{S_{PI}} \tag{39}$$

then the permissible reference plasticity diameter ζ_{zul} of the outer part is determined by solving the transcendent Equation (32) using

$$p = \frac{p_{PA}}{S_{PA}} \tag{40}$$

Calculation is then carried out for cases 1 and 2 both together. For the outer part to be subjected to elastic-plastic stresses and the actual safety against fully plastic stresses to be at least equal to the desired safety for the inner part in case 1 and for the outer part in case 2, the reference effective interference, ξ_w, shall be within the range according to Equation (41).

$$\frac{2}{\sqrt{3}}\frac{R_{eLA}}{E} < \xi_w < \frac{2}{\sqrt{3}}\zeta_{zul}^2\frac{R_{eLA}}{E} \tag{41}$$

The left term in Equation (41) is the condition for elastic-plastic stresses of the outer part. If $\xi_w < 2\,R_{eLA}/(\sqrt{3}\,E)$, then the outer part is being subjected to purely elastic stresses, and 4.2.3 shall be applied.

The reference plasticity diameter of the outer part, ζ, that results from the reference effective interference, ξ_w, is

$$\zeta = \sqrt{\frac{\sqrt{3}\,\xi_w E}{2\,R_{eLA}}} = 0{,}931\sqrt{\frac{\xi_w E}{R_{eLA}}} \tag{42}$$

The relevant joint pressure is calculated as follows:

$$p = \frac{R_{eLA}}{\sqrt{3}}[1 + 2\ln\zeta - (Q_A\zeta)^2] \tag{43}$$

Finally it shall be verified that the condition

$$\frac{q_{PA}}{q_A} \leq 0{,}3$$

(cf. Equation (35)) is met and the appropriate empirical value is taken into account (see also [21]).

It is also possible to combine a purely elastic design (e.g. in the case of a maximum fit) with an elastic-plastic design (for example, in case of a minimum fit). The design for elastic stress shall be according to 4.2, and for elastic-plastic stress, according to 4.3, of this standard.

5 Coefficients of adhesion for interference fits

5.1 General

The coefficient of adhesion ν_l is the ratio of the measured separation force to the calculated normal force in the joint. The coefficient of adhesion ν_r is the ratio of the measured sliding force to the calculated normal force in the joint. The coefficients of adhesion ν shall be distinguished from the (adhesive or sliding) friction coefficients, in which the measured normal force occurs instead of the calculated normal force. The coefficients of adhesion differ depending on whether the separation force or sliding force is acting in the longitudinal or circumferential direction (see also [22]). The coefficients of adhesion are identified by two indices (see Table 2).

Table 2 — Identification of coefficients of adhesion

	Separation	Sliding
Circumferential direction	ν_{lu}	ν_{ru}
Longitudinal direction	ν_{ll}	ν_{rl}

It is necessary to distinguish whether the coefficients of adhesion are determined for force fits or shrink fits. Coefficients of adhesion depend on the following influencing variables:

— the material combination of the inner and outer parts,

— the roughness of the joint surfaces,

— the state of the joint surfaces prior to joining (lubricated/unlubricated),

— the stress in the inner and outer parts (purely elastic or elastic-plastic).

As a result of friction in the joint, the coefficients of adhesion are subjected to statistical deviations. Thus, the coefficients of adhesion specified in the following tables are guideline values. These are conservative. They can be replaced by values determined empirically.

In the case of coefficients of adhesion, it is also necessary to distinguish between determinations made for a steady load and those made for a oscillating load. For coefficients of adhesion for oscillating load, see [5] and [16].

5.2 Coefficients of adhesion for force fits

The values given in Table 3 apply to steady loading. They are determined for the inner parts, on the basis of X 210 Cr W12 steel (name according to DIN EN ISO 4957). After joining, the interference fits should be left for about 24 hours before the load is first applied.

5.3 Coefficients of adhesion for shrink fits

Table 4 gives conservative coefficients of adhesion for shrink fits. They can be applied to sliding in the circumferential direction as well as in the longitudinal direction.

6 Design of interference fits

6.1 General

General rules for designing interference fits shall be applied to all interference fits. In addition, there are specific design rules for interference fits that are subjected to oscillating bending and/or torsional moments.

6.2 General design rules

— For interference fits in blind holes, it is necessary to provide ventilation (Figure 4).

— To enable axial joining, positional limits necessitated by the design or production technology are to be provided.

— In order to transmit high torques or axial forces, a full inner part should be paired with an outer part that is not too thin-walled ($Q_A \leq 0{,}5$). This, however, only applies to shaft-hub connections and not to rims.

— With sufficiently ductile materials (see 4.1 and, particularly conditions (8) and (9)), the joint pressure, p, and with it, the transferable circumferential and axial forces, can be increased by an elastic-plastic design (see 4.3). The greatest possible gain in joint pressure, p, in comparison with the purely elastic design is in the range $0{,}3 \leq Q_A \leq 0{,}4$.

Table 3 — Coefficients of adhesion for force fits with steady loading [6]

Materials			Coefficients of adhesion			
old	new		dry		lubricated	
		Number	ν_{ll}	ν_{rl}	ν_{ll}	ν_{rl}
St 60-2	E 335	1.0060	0,11	0,08	0,08	0,07
RSt37-2	S 235JRG2	1.0038	0,10	0,09	0,07	0,06
GS-60	GE 300	1.0558	0,11	0,08	0,08	0,07
GG-25	EN-GJL-250	0.6025	0,12	0,11	0,06	0,05
GGG-60	EN-GJS-600-3	0.7060	0,10	0,09	0,06	0,05
G-AlSi12(Cu)	EN AB-44000 ff.		0,07	0,06	0,05	0,04
G-CuPb10Sn (G-CuSn10Pb10)	CB495K	2.1176.01	0,07	0,06	— [a]	— [a]
TiAl6V4	TiAl6V4	3.7165.10	— [a]	— [a]	0,05	— [a]

[a] Coefficients are not known.

Table 4 — Coefficients of adhesion for shrink fits in longitudinal and circumferential direction applicable to sliding

Material combination, lubrication, joining	Coefficients of adhesion ν_r, ν_{rl}, ν_{ru}
Combination of steel-steel	
Pressurized oil assembly, joined with mineral oil	0,12
Pressurized oil assembly with degreased contact surfaces, joined with glycerine	0,18
Normal shrink fit after heating the outer part up to 300 °C in an electric oven	0,14
Shrink fit with degreased contact surfaces, after heating the outer part up to 300 °C in an electric oven	0,20
Combination of steel-cast iron	
Pressurized oil assembly, joined with mineral oil	0,10
Pressurized oil assembly with degreased contact surfaces	0,16
Combination of steel-MgAl, dry	0,10 to 0,15
Combination of steel-CuZn, dry	0,17 to 0,25

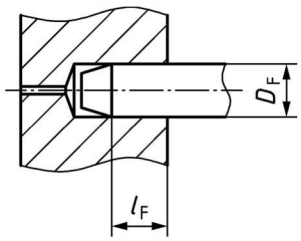


Figure 4 — Example of an air vent in blind holes

6.3 Design rules for interference fits with oscillating loads

In service, interference fits are often subject to cyclic or pulsating torsion and/or circumferential bending. The oscillating moments can cause sliding movements (slip) in the joint that change directions (see [1] or [15]). According to known studies on fracture due to prolonged friction [7], the durability of frictionally paired components is greatly reduced as slip increases. This slip can only be avoided if the joint pressure does not at any point fall below a threshold value determined by friction coefficients and shear stress in the plane of the joint. For interference fits with oscillating loads, load limits have been determined in [1] and [15] at which a slip path occurring at the hub edge is harmless, but only just. Thus it is evident that, in contrast to the ultimate bending moment, the ultimate torsional moment practically no longer increases above $l_F/D_F = 0{,}5$.

Interference fits shall be designed in such a way that the stresses occurring in connection with the subsisting operating conditions can be withstood [18]. Damage induced by notch effects, especially in the edge zone of interference fits, shall be taken into account in the fatigue analysis and if necessary, shall be reduced to a minimum by taking appropriate design measures [19], [20].

An optimal design of interference fits predominantly subjected to bending is achieved if a design as shown in Figure 5 is selected [5], [8]. The design should roughly employ the following geometrical relationships:

$$D_F/D_W \approx 1{,}1 \qquad r/(D_F - D_W) \approx 2 \tag{44}$$

On the other hand, any measures in the hub (recess) intended to reduce the joint pressure are difficult to assess, because the joint pressure at the hub edge may fall below the relevant limit value if the hub geometry is not correct. Despite this, if the shaft-hub connection is calculated accurately, for example using finite element calculations, it is possible to ensure a reduction in the joint pressure and increase in safety while also ensuring the joint pressure is sufficient to prevent fretting corrosion.

Unfortunately, for such shaft-hub connections, it is not possible to specify simple design guidelines as in Equation (44). The compromise between the measure for increasing safety by reducing the joint pressure and securing a sufficiently high joint pressure to prevent fretting corrosion can, in this case, only be found by finite element calculation.

If an inner part as shown in Figure 5 cannot be provided with a shoulder, the design shown in Figure 6 can also be used. For the notch radius, conditions analogous to Equation (44) are applicable.

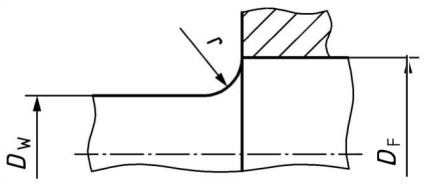


Figure 5 — Optimized interference fit on shaft shoulder

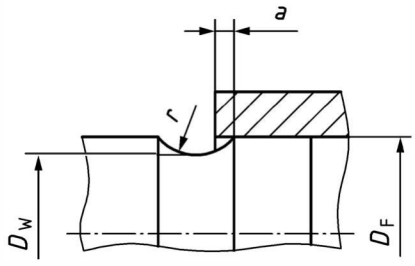


Figure 6 — Interference fit optimized by means of relief radius and axial projection

An axial projection *a*, as in Figure 6, results in a stiffening of the hub and thus in a higher joint pressure at the hub edge. This is an effective method of preventing fretting corrosion without simultaneously having to increase the interference. For the axial projection *a*, the following is valid for both the change in diameter (Figure 5) and the relief radius (Figure 6):

$$a \geq 0 \tag{45}$$

This tolerancing of the axial dimension *a*, shall ensure that this condition is met. An excessively large projection results in a decrease in fatigue strength. If $a < 0$, this is disadvantageous, because this results in stress ratios that are once again closer to those of a shaft that is not provided with a shoulder (see Figure 2).

The following design guidelines apply in particular:

a) Stresses due to cyclic or pulsating torque:

— the coefficient of adhesion in the joint surface shall be kept as high as possible, e.g. in shrink fits, by degreasing before assembly;

— the joint pressure, *p,* shall be as large as possible (in some cases this can be achieved by an elastic-plastic design, cf. 4.3). This is especially applicable at the entry of the inner part that transfers the torsion to the outer part. At this position, the measures that reduce the joint pressure p may be detrimental (see Figure 7);

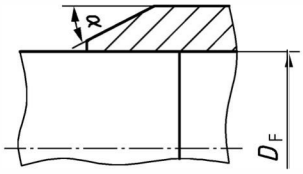


Figure 7 — Outer part with conical run-out

— if the inner and the outer parts are made of materials with different elastic constants, then the inner part should have the greater modulus of elasticity ($E_I > E_A$). The sliding moment of the interference fit should be, at the most, equal to that of the torsional moment at which plastic deformations occur in the free inner part (outside the outer part).

Generally, a reference joint length is sufficient for this:

$$l_F/D_F \leq 1{,}5 \tag{46}$$

b) Cyclic or rotating bending moments:

— to prevent axial drifting of the shaft, the following shall apply

$$l_F/D_F > 0{,}5 \tag{47}$$

— solid inner parts are more favourable than hollow ones. Where possible, the latter shall be avoided where there are large cyclic or rotating bending moments.

7 General information in technical documents

The manufacturing drawing shall contain the information required for the production of interference fits. Unless agreed upon otherwise, the “hole-basis fit system” (according to DIN EN ISO 286-1) is applicable.

The following tolerance intervals are recommended:

— for holes up to 500 mm diameter:
H7;

— for holes above 500 mm diameter:
H8;

— for shafts up to 500 mm diameter:
tolerance interval of tolerance grade IT6;

— for shafts above 500 mm diameter:
tolerance interval of tolerance grade IT7.

For the surface roughness of the joint surfaces, the following guideline values apply:

— for holes up to 500 mm diameter:
$Ra = 1{,}6$ µm;

— for holes above 500 mm diameter:
$Ra = 3{,}2$ µm;

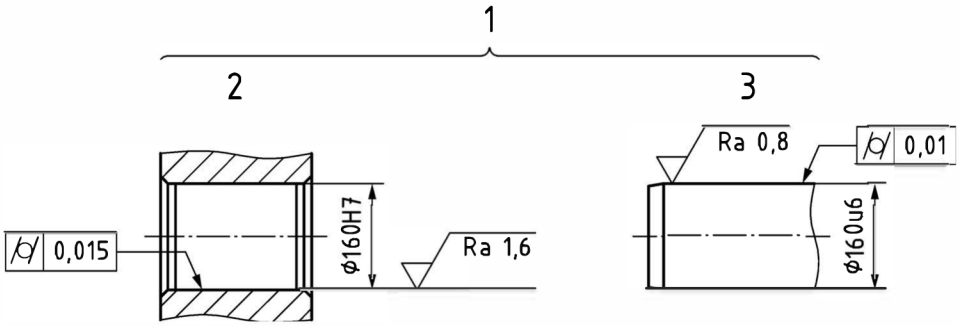
— for shafts up to 500 mm diameter:
$Ra = 0{,}8$ µm;

— for shafts above 500 mm diameter:
$Ra = 1{,}6$ µm.

The cylindrical form tolerance is about 1/3 of the dimensional tolerance for the diameter of the hub and the diameter of the shaft.

In the manufacturing drawing, the roughness values of the joint surfaces, generally the roughness average, *Ra*, and the tolerances of form (see DIN EN ISO 1101) shall be specified.

Examples of dimensions and tolerances of the parts to be joined:



Key

1 shrink fit
2 outer part
3 inner part

Figure 8 — Dimensioning and tolerancing examples

Indication of the toleranced dimensions and limit deviations on the drawing (if necessary)

Table 5 — Tolerances

Toleranced dimension	Limit deviations
⌀ 160 u6	+0,215 +0,190
⌀ 160 H7	+0,040 0

Examples how to indicate the joint temperature and cleanliness of the joint surfaces:

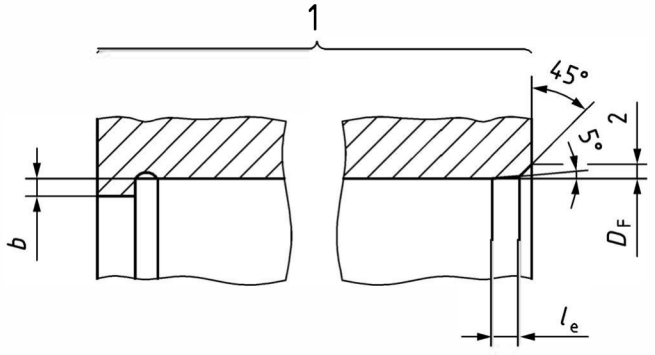
Outer part

Part ..... uniformly heated.
Heating temperature 280 °C
at approximately 20 °C room temperature
Joint surfaces free of lubricant and foreign matter

Inner part (only when supercooling is required)

Part ..... uniformly supercooled.
Supercooling temperature −110 °C
at approximately 20 °C room temperature
Joint surfaces free of lubricant and foreign matter

For rims (e.g. gear rims, wheel rims, etc.), the design suitable for joining is shown in Figure 9. The height of the bore shoulder, b, shall ensure an axial fixing.



Key

1 outer part
b bore shoulder

Figure 9 — Design of outer parts suitable for joining, using a rim as an example; l_e according to Equation (49)

8 Assembly of interference fits

8.1 Assembly of force fits

The required press-in force is as follows

$$F_e = \pi D_F l_F \nu_{ll} p_{max} \quad (48)$$

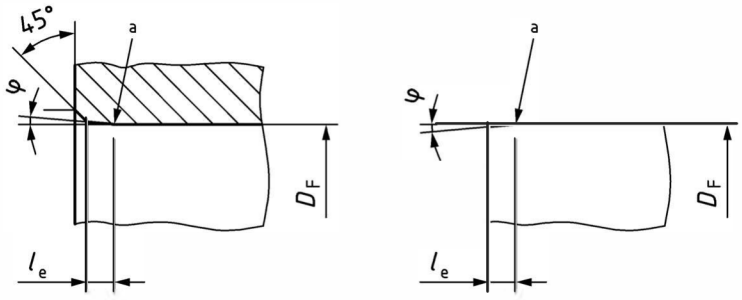
The coefficient of adhesion ν_{ll} can be taken from Table 3. The maximum joint pressure p_{max} shall be calculated for the maximum interference U_g.

Omitting to lubricate the joint surfaces before joining results in higher coefficients of adhesion and thus higher transferable longitudinal and circumferential forces. However, there is a risk of fretting on unlubricated joint surfaces, especially in the case of elastic-plastic design. For this reason, the joint surfaces shall be lightly oiled before assembly.

Furthermore, the following information regarding the design shall be taken into account (see Figure 10):

— There shall be no sharp edges or transitions on the parts to be joined.

— The chamfer angle, φ, should not exceed 5°.



Key

a rounded transition

Figure 10 — Design of force fits; l_e according to Equation (49)

The values for the length of the chamfer, l_e, are given in Table 6.

Table 6 — Chamfer length, l_e

D_F		l_e	D_F		l_e
over	up to		over	up to	
50	80	4	400	630	8
80	160	5	630	800	9
160	250	6	800	1000	10
250	400	7	1000	—	10

— For the calculation of the chamfer length, the following applies:

$$l_e \approx \sqrt[3]{D_F} \tag{49}$$

— The press-in chamfer shall be applied to the assembly counterpart with the higher yield strength (usually on the inner part).

— Long, slender inner parts shall be checked for buckling.

8.2 Thermal joining of shrink fits

Shrink fits are joined by supercooling the inner part or by heating the outer part. Where interferences are relatively large, the two methods are combined. The calculation of the temperatures to which the inner part is supercooled or the outer part is heated is based on the selected minimum fit. For joining, a joining clearance shall also be maintained in order to exclude the possibility of adhesion of the interference fit during the joining process.

For one-off production (i.e. of a single unit), it is recommended using a joining clearance of

$$U_{s\vartheta} = 0{,}001\, D_F \tag{50}$$

Equation (50) covers the risk of premature adhesion of the joining partners before the joining process of a single unit is complete. When using equipment for the joining process, a lower clearance than determined according to Equation (50) can be used.

The following applies for the interference for joining purposes:

$$U_F = U_g + U_{s\vartheta} \tag{51}$$

The room temperature, ϑ_R, and the joining temperature, ϑ_I, of the inner part have generally been defined. The required joining temperature of the outer part is calculated as follows:

$$\vartheta_{Aerf} = \vartheta_R + \frac{U_F}{\alpha_A\, D_F} + \frac{\alpha_I}{\alpha_A}(\vartheta_I - \vartheta_R) \tag{52}$$

The highest occurring joining temperature shall not interfere with the required properties of the heat-treated parts that are to be joined.

The coefficients of thermal linear expansion for the inner and outer parts are given in Table 7.

For supercooling the elongation fits, either CO_2 dry ice ($\vartheta_I = -78{,}4$ °C) or liquid nitrogen ($\vartheta_I = -195{,}8$ °C) can be used. The decision as to whether complex supercooling is required shall be based on the maximum permissible temperature of the outer part.

Depending on the material of the outer part and the heat treatment, the specifications in Table 8 are applicable for the maximum joining temperatures.

For terms and definitions relating to the heat treatment of ferrous materials, see DIN EN 10052.

The permissible joining temperature, ϑ_{Azul}, of the outer part shall be specified in the assembly drawing. For the maximum joinable interference, the following applies:

$$U_{max} = D_F[\alpha_A(\vartheta_{Azul} - \vartheta_R) - \alpha_I(\vartheta_I - \vartheta_R)] - U_{s\vartheta} \tag{53}$$

9 Instructions for making interference fits

9.1 General information

The dimensions and the tolerances given in the drawing, and all other special information, shall be ensured by carrying out checks during the manufacturing process. The test reports of individual checks allow the pairing of joining parts that match each other, and these shall be marked to prevent false pairings. The joint surfaces shall be checked to ensure they are free of dirt and corrosion. The appropriate safety regulations are to be followed.

9.2 Making interference fits by pressing in

Before pressing in, a thin layer of oil shall be applied to the entire surface of the joint surfaces. The use of additives such as molybdenum disulphide is permissible only if stated in the working documents. The joining parts shall not move out of place during the press-fitting process. Stick-slip can be avoided by using press-in or press-out speeds of about 50 mm/s and by ensuring adequate compressive force reserves are available (about 2,5 times the separation force). Interference fits should be left for at least 24 hours before being placed under load.

Table 7 — Poisson's ratio, elastic modulus, coefficient of linear expansion

Material designation/ no. according to DIN	Material designation/ no. according to EN	Poisson's ratio μ	Elastic modulus N/mm²	Coefficient of linear expansion α $\frac{10^{-6}}{°C}$	
				Heating	Supercooling
MgAl8Znn / 3.5812 AlMgSi / 3.2315 AlCuMg / 3.1325	EN-MCMgAl8Zn1 / EN-MC21110 EN AW-Al Si1MgMn / EN AW-6082 EN AW-Al Cu4MgSi(A) / EN AW-2017A	0,3 0,34 0,33 to 0,34	65 000 to 75 000	23	18
GG-10 / 0.6010 GG-15 / 0.6015	EN-GJL-100 / 5.1100 [a] EN-GJL-150 / 5.1200 [a]	0,24	70 000 80 000	10	8
GG-20 / 0.6020 GG-25 / 0.6025	EN-GJL-200 / 5.1300 [a] EN-GJL-250 / 5.1301 [a]	0,24 to 0,26	105 000 130 000		
GGG-50 / 0.7050	EN-GJS-500-7 / 5.3200	0,28 to 0,29	> 140 000	10	8
Malleable cast iron		0,25	90 000 to 100 000	10	8
C-steel low alloyed Ni-steel		0,3 to 0,31 0,31	200 000 to 235 000	11	8,5
Bronze		0,35	80 000 to 85 000	16	14
Red casting brass		0,35 to 0,36		17	15
CuZn39Pb3 / 2.0401 CuZn37 / 2.0321	CuZn39Pb3 / CW614N CuZn37 / CW508L	0,37 0,36		18	16

[a] Not approved for industrial plant engineering in metallurgical plants and rolling mills.

Table 8 — Joining temperature

Material of the outer part (Hub)	Joining temperature °C max
Low strength construction steel Cast steel Cast iron with spherical graphite	350
Tempered steel or cast steel	300
Surface-hardened steel	250
Case-hardened steel or high tempered structural steel	200

9.3 Making shrink fits

9.3.1 General

Before the start, auxiliary materials shall be checked for absolute operational reliability. Handling equipment (e.g. cranes) that has a rapid lowering system are to be preferred. The clearance is reduced by heat dissipation during shrinkage or heat absorption during elongation. In order to prevent the parts becoming fixed prematurely, the joining process shall be carried out as quickly as possible in a draught-free room. It is very important that the heating and supercooling process takes place steadily and is fully completed.

9.3.2 Heating the outer part

Local overheating should be prevented. Different heating options are listed in Table 9, together with relevant remarks. Parts that cannot withstand the heating temperature, such as radial shaft seals, shall be temporarily removed. The cooling of the joined parts should take place as steadily and as slowly as possible (use a heat-insulating substrate, and if possible, cover the fit).

9.3.3 Supercooling the inner part

Supercooling of the inner part is mostly used for serial parts that have already been finished. Care shall be taken to ensure that transport distances are extremely short in cases where the outer part is additionally heated. A sufficient quantity of coolant shall be made available. The losses calculated for this case can be assumed to be lower if the cooling tank is of just sufficient capacity, the work piece is fully immersed, and the small parts are the last to be supercooled in the same container. Table 10 lists substances suitable for supercooling, together with relevant remarks.

9.3.4 Hydraulic joining and separation

Pressure generators, supply lines, connecting holes and distribution grooves shall be clean and the joint surfaces shall be undamaged (i.e. there shall be no scratches, whether surface scratches or the deeper type, in the longitudinal direction). The pressure generators used shall be able to withstand a surface pressure 2 to 5 times that which has been calculated. It is advantageous to use lighter oils for joining and heavier oils for separating; glycerine shall be used at very high pressures. Hydraulically joined fits are allowed to be subjected to loading only after degradation of the oil film (10 min to 2 h).

Separation:

As a precaution, a rust remover can be added to the pressurizing medium. Fits with conical joints shall be axially secured. Where expansion due to pressure puts parts at risk, it is necessary to check the joint pressure using pressure gauges (friction pointers).

Table 9 — Means of heating

Means of heating	Used for	Remarks
Electric heating plates	(Mostly small) parts produced in series	Heating often incomplete, risk of local overheating!
Electric heating cores	Cases and hubs	Attainable joining temperature: up to ≈ 50 °C
Annular burner	(Mostly bulky) outer parts, if inner part is supercooled	Often multiple burners (surface heaters) required simultaneously. Risk of distortion, warpage and local overheating
Immersed heating	Outer parts on whose joint surfaces oil may be present during assembly	Natural organic heat transfer medium up to 300 °C; paraffin and silicon-based oils up to 400 °C
Hot air oven and hot air chamber	Outer parts whose joint surfaces are required to be dry and free from oxide layers	Usually up to 400 °C heating temperature; in special ovens temperatures up to 650 °C possible
Gas-heated oven	All outer parts whose joint surfaces are required to be dry and where any oxide layers that increase the coefficient of adhesion are either intentional or of no consequence	Excess oxide or scale-like coatings on the joint surfaces can be avoided by making special provisions (using a lockable muffle or gas flame protection).
Electric oven		

Table 10 — Substances for supercooling

Substances for supercooling	Chemical formula	Boiling point of the gas	Comments
Carbon dioxide snow or dry ice	CO_2	−78,4 °C	Part to be joined is cooled relatively slowly; cooling is faster if alcohol is used as the heat carrier. Addition of trichloroethylene prevents freezing of the surfaces of the parts to be joined
Liquid nitrogen	N_2	−195,8 °C	When used in enclosed rooms, provide plenty of ventilation. Otherwise, there are no particular risks
The use of liquefied oxygen or liquefied air is not recommended due to the high risk of explosion.			

10 Special information

10.1 Fatigue strength analysis

With interference fits that transfer oscillating torsion and/or bending moments, the inner part is subjected to fatigue. For this reason, a fatigue strength analysis shall be carried out according to DIN 743-1 to DIN 743-3.

10.2 Stresses due to centrifugal force

If an interference fit is subjected to centrifugal force in addition to the stresses caused by joining, the following shall be taken into account from the aspect of safety. Due to the centrifugal force, the inner and the outer parts expand in radial direction, with the outer part undergoing greater elongation than the inner part. This results in a reduction of the joint pressure p compared with the pressure when the fit is at rest. However, along with the joint pressure p the transferable torque and the transferable axial force are also reduced (see Equations (1) and (2)). In extreme cases, complete loss of joint pressure p can result in the outer part lifting off from the inner part.

The designs below apply under the following conditions:

a) The inner part is solid ($Q_\mathrm{I} = 0$).

b) The inner and outer parts have equal elasticity constants ($E_\mathrm{I} = E_\mathrm{A} = E$; $\mu_\mathrm{I} = \mu_\mathrm{A} = \mu$). Furthermore, their densities are equal ($\rho_\mathrm{I} = \rho_\mathrm{A} = \rho$).

NOTE $\mathrm{kg/dm^3} \approx 10^{-9}\,\mathrm{Ns^2/mm^4}$

c) As a result of joining and the centrifugal force, only elastic stresses occur in the inner and the outer part.

For condition c) to be met, the reference effective interference, ξ_w, shall satisfy the following inequations [9]:

$$\xi_\mathrm{w} \leq \frac{3+\mu}{3+\mu+(1-\mu)\,Q_\mathrm{A}^2}\,\frac{R_\mathrm{eLA}}{E} \tag{54}$$

$$\xi_\mathrm{w} \leq \frac{R_\mathrm{eLI}}{Q_\mathrm{A}^2\,E} \tag{55}$$

If the interference fit rotates with a speed n, then the peripheral speed u of the outer contour of the outer part is

$$u = \pi\, n\, D_\mathrm{Aa} \tag{56}$$

The joint pressure, p_n, at a rotational speed n is calculated from the joint pressure p at rest according to

$$p_n = \left[1 - \left(\frac{u}{u_\mathrm{ab}}\right)^2\right] p \tag{57}$$

The joint pressure, p, at rest is calculated according to Equation (17). The peripheral speed u_ab at which the outer part lifts off from the inner part is given by

$$u_\mathrm{ab} = 2\sqrt{\frac{E\,\xi_\mathrm{w}}{(3+\mu)\rho}} \tag{58}$$

If a drop in the joint pressure caused by the centrifugal force is to be kept at a maximum of 10 % compared to the pressure when the assembly is at rest, then the peripheral speed of the outer contour of the outer part shall meet the condition

$$u \leq 2\sqrt{0{,}1}\sqrt{\frac{E\,\xi_\mathrm{w}}{(3+\mu)\rho}} = 0{,}632\sqrt{\frac{E\,\xi_\mathrm{w}}{(3+\mu)\rho}} = 0{,}316\;u_\mathrm{ab} \tag{59}$$

For interference fits subjected to purely elastic stresses that do not meet conditions a) and/or b), the calculation methods are given in [9]. For interference fits subjected to elastic-plastic stresses, reference is made to [10].

In a purely elastic design it shall be ensured that when the interference fit is at its maximum rotational speed, the joint pressure according to Equation (57) is sufficient for the transfer of the circumferential and/or axial force. An adequate safety margin against the rotational speed at which the interference fit lifts off shall be maintained at all events. When the outer part lifts off, the interference fit not only loses its function as a frictional connection, but radial drifting of the outer part can also result in great imbalances and a risk to the entire unit.

10.3 Inspection certificate

In application cases in which injury to persons cannot be ruled out if the interference fit fails, the prescribed tests (in the form of acceptance testing) shall be performed.

The inspection certificate is an integral part of the package delivered to the customer (see Annex C for an example).

10.4 Marking

This marking is only required if an inspection certificate according to 10.3 is to be issued. The information given by the workpiece marking shall contain the same information as the inspection certificate.

The following information shall be marked on the front so as to be clearly and permanently visible:

1st line: manufacturer's logo

2nd line: manufacturer's order no., drawing no., item no.

3rd line: customer part number (only for spare parts)

Other requirements and manufacturing requirements:

In interference fits in which the outer part is a rim, gear rim or wheel rim, for example, a permanent radial marking shall be applied by the manufacturer on the front surface of the joint, near the marking. This enables any displacement, e.g. micro-wandering of the gear wheel, to be detected during an inspection. If required, verification should be carried out by computation according to the information in [12] and [17] in order to prevent micro-wandering in gear rims that have undergone shrinkage.

Interference fits shall be free of any cracks and internal tears that would affect usability.

When producing an interference fit, it shall be ensured that the technical requirements according to the information in the manufacturing drawing are met. The specifications established in the calculations (e.g. interferences, coefficients of adhesion as a function of the material combination, lubrication and joining, as well as a uniform joining temperature of the outer part (hub, gear rim, rim) and inner part (shaft)), shall be reliably maintained during manufacturing.

The equipment required for producing the fit, e.g. the means of heating, the substance used for supercooling, and the measuring and testing devices shall be shown to be available, and it shall also be shown that the joining temperature has been verified [13], [14].

The manufacturer shall ensure that only skilled persons are permitted to perform the production process; and that only experts shall be responsible for the testing and production control of interference fits.

Annex A
(informative)

Calculation examples

A.1 Purely elastic design of an interference fit for a given joint pressure, p

Given: $D_F = 50$ mm $D_{il} = 0$ $D_{aA} = 100$ mm

$Rz_I = 6$ µm $Rz_A = 6$ µm

$R_{eLI} = 200$ N/mm2 $R_{eLA} = 300$ N/mm^2

$E_I = E_A = E = 215\,000$ N/mm^2

$\mu_I = \mu_A = \mu = 0{,}3$

$p = 90$ N/mm^2 $S_p = 1{,}1$

Sought: required interference U

Calculated values

According to Equations (10) and (11): $Q_A = \frac{50}{100} = 0{,}5$ $Q_I = 0$

Verification of Equation (15): $90\ \text{N/mm}^2 < \frac{1 - 0{,}5^2}{\sqrt{3} \cdot 1{,}1} \cdot 300\ \text{N/mm}^2 = 118{,}1\ \text{N/mm}^2$

The requirement according to Equation (15) is met, consequently the outer part is subjected to purely elastic stresses.

Verification of Equation (17): $90\ \text{N/mm}^2 < \frac{2 \cdot 200}{\sqrt{3} \cdot 1{,}1}\ \text{N/mm}^2 = 210\ \text{N/mm}^2$

The requirement according to Equation (17) is met, the solid inner part is subjected to purely elastic stresses.

According to Equation (19): $\xi_w = \frac{2}{1 - 0{,}5^2} \cdot \frac{90}{215\,000} = 1{,}116 \cdot 10^{-3}$

According to Equation (12): $U_w = 50 \cdot 1{,}116 \cdot 10^{-3}$ mm $= 55{,}8 \cdot 10^{-3}$ mm $U_w = 56$ µm

According to Equation (7): $U = 56\ \mu\text{m} + 0{,}4 \cdot (6 + 6)\ \mu\text{m} = 60{,}8\ \mu\text{m}$ $U = 61$ µm

A.2 Purely elastic design of an interference fit for a given interference, U

Geometric and material data as in Example A.1

Given: $U = 75$ µm

Calculated values

According to Equation (7): $U_w = 75\ \mu\text{m} - 0{,}4 \cdot (6 + 6)\ \mu\text{m} = 70{,}2\ \mu\text{m} = 70\ \mu\text{m}$

According to Equation (12): $\xi_w = \frac{70 \cdot 10^{-3}}{50} = 1{,}4 \cdot 10^{-3}$

Verification of Equation (22) $1{,}4 \cdot 10^{-3} < \dfrac{2 \cdot 300}{\sqrt{3} \cdot 1{,}1 \cdot 215\,000} = 1{,}465 \cdot 10^{-3}$

The requirement according to Equation (22) is met, consequently the outer part is subjected to purely elastic stresses.

Verification of Equation (23): $1{,}4 \cdot 10^{-3} < \dfrac{4 \cdot 200}{\sqrt{3} \cdot (1 - 0{,}5^2) \cdot 1{,}1 \cdot 215\,000} = 2{,}6 \cdot 10^{-3}$

The requirement according to Equation (23) is met, consequently the solid inner part is subjected to purely elastic stresses.

According to Equation (25): $p = \dfrac{1 - 0{,}5^2}{2} \cdot 215\,000 \cdot 1{,}4 \cdot 10^{-3}\ \text{N/mm}^2 = 112{,}9\ \text{N/mm}^2$

A.3 Purely elastic design of an interference fit for a given joint pressure, *p*

Given: $D_{\text{iI}} = 10$ mm; $D_{\text{F}} = 50$ mm; $D_{\text{aA}} = 125$ mm

$Rz_{\text{I}} = 6$ µm; $Rz_{\text{A}} = 6$ µm

$R_{\text{eLI}} = 300\ \text{N/mm}^2$; $R_{\text{eLA}} = 150\ \text{N/mm}^2$

$E_{\text{I}} = 215\,000\ \text{N/mm}^2$; $E_{\text{A}} = 70\,000\ \text{N/mm}^2$

$\mu_{\text{I}} = \mu_{\text{A}} = \mu = 0{,}3$

$p = 50\ \text{N/mm}^2$; $S_{\text{P}} = 1{,}2$

Sought: required interference *U* for a purely elastic design

Calculated values

According to Equations (10) und (11): $Q_{\text{A}} = \dfrac{50}{125} = 0{,}4$ $\quad Q_{\text{I}} = \dfrac{10}{50} = 0{,}2$

Verification of Equation (15): $50\ \text{N/mm}^2 < \dfrac{1 - 0{,}4^2}{\sqrt{3} \cdot 1{,}2} \cdot 150\ \text{N/mm}^2 = 60{,}6\ \text{N/mm}^2$

The requirement according to Equation (15) is met, consequently the outer part is subjected to purely elastic stresses.

Verification of Equation (16): $50\ \text{N/mm}^2 < \dfrac{1 - 0{,}2^2}{\sqrt{3} \cdot 1{,}2} \cdot 300\ \text{N/mm}^2 = 138{,}6\ \text{N/mm}^2$

The requirement according to Equation (16) is met, consequently the inner part is subjected to purely elastic stresses.

According to Equation (13): $K = \dfrac{70\,000}{215\,000} \cdot \left(\dfrac{1 + 0{,}2^2}{1 - 0{,}2^2} - 0{,}3 \right) + \dfrac{1 + 0{,}4^2}{1 - 0{,}4^2} + 0{,}3 = 1{,}936$

According to Equation (18): $\xi_{\text{w}} = 1{,}936 \cdot \dfrac{50}{70\,000} = 1{,}383 \cdot 10^{-3}$

According to Equation (12): $U_{\text{w}} = 1{,}383 \cdot 10^{-3} \cdot 50\ \text{mm} = 69{,}1 \cdot 10^{-3}$ $\quad U_{\text{w}} = 69$ µm

According to Equation (7): $U = 69\ \mu\text{m} + 0{,}4 \cdot (6 + 6)\ \mu\text{m} = 73{,}8\ \mu\text{m}$ $\quad U = 74$ µm

A.4 Purely elastic design of an interference fit for a given interference, *U*

Given: Geometric and material data as in Example A.3

$U = 65\,\mu m$

Sought: Joint pressure, p, and verification of purely elastic design

Calculated values

According to Equation (7): $U_w = 65\,\mu m - 0{,}4 \cdot (6 + 6)\,\mu m = 60{,}2\,\mu m$

According to Equation (12): $\xi_w = \frac{60{,}2 \cdot 10^{-3}}{50} = 1{,}204 \cdot 10^{-3}$

The auxiliary value K according to Equation (13) is taken from Example A.3, as the geometric and material data are unchanged.

Verification of Equation (20): $1{,}204 \cdot 10^{-3} < 1{,}936 \cdot \frac{1 - 0{,}4^2}{\sqrt{3} \cdot 1{,}2} \cdot \frac{150}{70\,000} = 1{,}676 \cdot 10^{-3}$

The requirement according to Equation (20) is met, consequently the outer part is subjected to purely elastic stresses.

Verification of Equation (21): $1{,}204 \cdot 10^{-3} < 1{,}936 \cdot \frac{1 - 0{,}2^2}{\sqrt{3} \cdot 1{,}2} \cdot \frac{300}{70\,000} = 3{,}832 \cdot 10^{-3}$

The requirement according to Equation (21) is met, consequently the inner part is subjected to purely elastic stresses.

According to Equation (24): $p = \frac{1{,}204 \cdot 10^{-3} \cdot 70\,000}{1{,}936}\ N/mm^2 = 43{,}5\ N/mm^2$

A.5 Elastic-plastic design of an interference fit for a given joint pressure, *p*

Given:

$D_{iI} = 0$ mm	$D_F = 50$ mm	$D_{aA} = 100$ mm
$Rz_I = 8\,\mu m$	$Rz_A = 8\,\mu m$	
$R_{eLI} = 400\ N/mm^2$	$R_{eLA} = 300\ N/mm^2$	
$E_I = E_A = E = 215\,000\ N/mm^2$		
$\mu_I = \mu_A = \mu = 0{,}3$		
$p = 220\ N/mm^2$	$S_{PI} = 1{,}1$	$S_{PA} = 1{,}25$

Sought: required interference U for elastic-plastic design

Calculated values

According to Equations (10) and (11): $Q_A = \frac{50}{100} = 0{,}5 \qquad Q_I = 0$

Calculation of the joint pressure limit p_{PA} according to Equation (28):

For $Q_A = 0{,}5 > 0{,}368$ gilt: $p_{PA} = -2 \cdot 300 \cdot \ln 0{,}5/\sqrt{3}\ N/mm^2 = 240{,}2\ N/mm^2$

but $220\ N/mm^2 > \frac{240{,}2}{1{,}25}\ N/mm^2 = 192{,}2\ N/mm^2$

The outer part provides inadequate safety against fully plastic stresses.

If only one material with $R_{eLA} = 300\ \text{N/mm}^2$ is available, then a design with $p = 220\ \text{N/mm}^2$ is not permissible due to inadequate safety against fully plastic stresses in the outer part.

Solution: Choice of material with $R_{eLA} = 400\ \text{N mm}^2$

Revised calculation of the joint pressure limit, p_{PA}, according to Equation (28):

$$p_{PA} = -2 \cdot 400 \cdot \ln 0{,}5/\sqrt{3}\ \text{N/mm}^2 = 320{,}2\ \text{N/mm}^2$$

Verification of Equation (26): $(1 - 0{,}5^2) \cdot 400/\sqrt{3}\ \text{N/mm}^2 = 173{,}2\ \text{N/mm}^2 < 220\ \text{N/mm}^2$

and $$220\ \text{N/mm}^2 < \frac{320{,}2}{1{,}25}\ \text{N/mm}^2 = 256{,}2\ \text{N/mm}^2$$

The requirement according to Equation (26) is met, consequently the outer part is subjected to elastic-plastic stresses with a sufficient safety margin against the fully plastic state.

Calculation of the joint pressure limit, p_{PI}, according to Equation (30):

$$p_{PI} = 2 \cdot 400/\sqrt{3}\ \text{N/mm}^2 = 462{,}0\ \text{N/mm}^2$$

Verification of Equation (29): $$220\ \text{N/mm}^2 < \frac{462{,}0}{1{,}1}\ \text{N/mm}^2 = 420{,}0\ \text{N/mm}^2$$

The requirement according to Equation (29) is met, consequently the inner part is subjected to purely elastic stresses. Determination of reference plasticity diameter, ζ, (see Annex B).

Table A.1 — Iteration steps for plasticity diameter, ζ

j	ζ_j	ζ_{j+1}	$\lvert\zeta_{j+1} - \zeta_j\rvert$
1	1	1,135 1	0,135 1
2	1,135 1	1,152 9	0,017 8
3	1,152 9	1,153 2	0,000 3

For further calculation: $\zeta = 1{,}153$

According to Equation (34): $$\xi_w = \frac{2}{\sqrt{3}} \cdot 1{,}153^2 \cdot \frac{400}{215\,000} = 2{,}858 \cdot 10^{-3}$$

According to Equation (12): $U_w = 50 \cdot 2{,}858 \cdot 10^{-3}\ \text{mm} = 0{,}142\,9\ \text{mm}$

$$U_w = 143\ \mu\text{m}$$

According to Equation (7): $U = 143\ \mu\text{m} + 0{,}4 \cdot (8 + 8)\ \mu\text{m} = 149{,}4\ \mu\text{m}$ $\quad U = 150\ \mu\text{m}$

Verification of the proportion of the plastic ring zone in relation to the overall cross section.

According to Equation (35): $$\frac{q_{PA}}{q_A} = \frac{(1{,}153^2 - 1) \cdot 0{,}5^2}{1 - 0{,}5^2} = 0{,}110 < 0{,}30$$

The plastic ring zone of the outer part is 0,11 of the overall cross section of the outer part and is permissible.

A.6 Recalculation of an elastic-plastic interference fit with a specified interference, *U*

Given: Geometric and material data as in Example A.5

however $R_{eLA} = 400\ \text{N/mm}^2$ $U = 190\ \mu\text{m}$

Sought: Joint pressure, p

According to Equation (7): $U_w = 190\ \mu\text{m} - 0{,}4 \cdot (8 + 8)\ \mu\text{m} = 183{,}6\ \mu\text{m}$ $U_w = 184\ \mu\text{m}$

According to Equation (12): $\xi_w = \frac{184 \cdot 10^{-3}}{50} = 3{,}68 \cdot 10^{-3}$

The test condition according to Equation (36) is not met:

$$(1 - 0{,}5^2) \cdot \frac{400}{2}\ \text{N/mm}^2 = 150\ \text{N/mm}^2 < 400\ \text{N/mm}^2$$

Elastic-plastic design of the interference fit is possible since, when subjected to joint pressure, the outer part changes from the purely elastic to the elastic-plastic state, and this is less than the joint pressure limit when the outer part is subjected to fully plastic stresses.

Determination of the permissible non-dimensional plasticity diameter, ζ_{zul}:

The values $p_{PI} = 462{,}0\ \text{N/mm}^2$ and $p_{PA} = 320{,}2\ \text{N/mm}^2$ are taken from Example 5.*)

Case 2 then applies: $\frac{320{,}2}{1{,}25}\ \text{N/mm}^2 = 256{,}2\ \text{N/mm}^2 < \frac{462{,}0}{1{,}1}\ \text{N/mm}^2 = 420\ \text{N/mm}^2$

According to Equation (40): $p = \frac{320{,}2}{1{,}25}\ \text{N/mm}^2 = 256{,}2\ \text{N/mm}^2$

With this value, the transcendental Equation (32) can be solved with the aid of the algorithm given in Annex B.

Table A.2 — Iteration steps for plasticity diameter, ζ

j	ζ_j	ζ_{j+1}	$\lvert\zeta_{j+1} - \zeta_j\rvert$
1	1	1,239 6	0,239 6
2	1,239 6	1,303 9	0,064 3
3	1,303 9	1,308 1	0,004 2
4	1,308 1	1,308 1	0

For further calculation: $\zeta_{zul} = 1{,}308$

Verification of Equation (41):

$$\frac{2}{\sqrt{3}} \cdot \frac{400}{215\,000} = 2{,}149 \cdot 10^{-3} < 3{,}68 \cdot 10^{-3} < \frac{2}{\sqrt{3}} \cdot 1{,}308^2 \cdot \frac{400}{215\,000} = 3{,}68 \cdot 10^{-3}$$

The outer part is subjected to elastic-plastic stresses to the limit of the fully plastic state.

*) Translator's note: Should read "A.5".

According to Equation (42): $\zeta = 0{,}931 \cdot \sqrt{\frac{3{,}68 \cdot 10^{-3} \cdot 215\,000}{400}} = 1{,}309$

According to Equation (43): $p = \frac{400}{\sqrt{3}} \cdot [1 + 2 \cdot \ln 1{,}309 - (0{,}5 \cdot 1{,}309)^2]\ \mathrm{N/mm^2} = 256{,}4\ \mathrm{N/mm^2}$

According to Equation (35): $\frac{q_{\mathrm{PA}}}{q_{\mathrm{A}}} = \frac{(1{,}309^2 - 1) \cdot 0{,}5^2}{1 - 0{,}5^2} = 0{,}237$

The plastic ring zone is 0,237 of the overall cross section of the outer part and is permitted.

A.7 Geometric and material data as in Example A.1

$$\rho_{\mathrm{I}} = \rho_{\mathrm{A}} = 7{,}85\ \mathrm{kg/dm^3} = 7{,}85 \cdot 10^{-9}\ \mathrm{Ns^2/mm^4}$$

$$n = 10\,000\ \mathrm{1/min} = 166{,}7\ \mathrm{1/s}$$

Verification of the effect of centrifugal force.

Conditions a) and b) according to 10.2 are met.

Verification of the validity of condition c):

Taken from Example A.1: $\xi_{\mathrm{w}} = 1{,}116 \cdot 10^{-3}$

From Equation (54):

$$\frac{3 + \mu}{3 + \mu + (1 - \mu)\, Q_{\mathrm{A}}^2} \frac{R_{\mathrm{eLA}}}{E} = \frac{3 + 0{,}3}{3 + 0{,}3 + (1 - 0{,}3) \cdot 0{,}5^2} \cdot \frac{300}{215\,000} = 1{,}325 \cdot 10^{-3}$$

$$\frac{R_{\mathrm{eLI}}}{Q_{\mathrm{A}}^2 E} = \frac{200}{0{,}5^2 \cdot 215\,000} = 3{,}721 \cdot 10^{-3}$$

Condition (54) is met and hence also the condition c).

From Equation (58), it follows that $u_{\mathrm{ab}} = 2 \cdot \sqrt{\frac{215\,000 \cdot 1{,}116 \cdot 10^{-3}}{(3 + 0{,}3) \cdot 7{,}85 \cdot 10^{-9}}} \frac{\mathrm{mm}}{\mathrm{s}} = 192{,}5 \cdot 10^3 \frac{\mathrm{mm}}{\mathrm{s}} = 192{,}5 \frac{\mathrm{m}}{\mathrm{s}}$

From Equation (56), it follows that $u = \pi \cdot 166{,}7 \cdot 100 \frac{\mathrm{mm}}{\mathrm{s}} = 52{,}4 \cdot 10^3 \frac{\mathrm{mm}}{\mathrm{s}} = 52{,}4 \frac{\mathrm{m}}{\mathrm{s}}$

From Equation (59), it follows that $u = 52{,}4 \frac{\mathrm{m}}{\mathrm{s}} < 0{,}316 \cdot 192{,}5 \frac{\mathrm{m}}{\mathrm{s}} = 60{,}8 \frac{\mathrm{m}}{\mathrm{s}}$

To which value does the join pressure $p = 90\ \mathrm{N/mm^2}$ drop at a rotational speed of 20 000 1/min?

From Equation (56), it follows that $u = \pi \cdot \frac{20\,000}{60} \cdot 100 \frac{\mathrm{mm}}{\mathrm{s}} = 104{,}70 \cdot 10^3 \frac{\mathrm{mm}}{\mathrm{s}} = 104{,}7 \frac{\mathrm{m}}{\mathrm{s}}$

From Equation (57), it follows that $p_n = \left[1 - \left(\frac{104{,}7}{192{,}5}\right)^2\right] \cdot 90\ \mathrm{N/mm^2} = 63{,}4\ \mathrm{N/mm^2}$

Annex B
(informative)

Procedure for the iterative calculation of the non-dimensional plasticity diameter ζ

The iteration rule is: $$\zeta_{j+1} = \zeta_j \left\{1 - \frac{2\ln\zeta_j - (Q_A\zeta_j)^2 + 1 - 1{,}732p/R_{eLA}}{2[1-(Q_A\zeta_j)^2]}\right\}$$

The start value is set as: $\zeta_1 = 1$

The iteration is discontinued if the condition

$$|\zeta_{j+1} - \zeta_j| < 0{,}001$$

is met.

The calculated relative plasticity diameter $\zeta = \zeta_{j+1}$ shall lie within the range $1 \leq \zeta \leq 1/Q_A$.

Annex C
(informative)

Example of an inspection certificate

Users of this standard may reproduce the forms in Annexes A to E without prejudice to the rights of DIN regarding the rest of the document.

Inspection certificate Summary for interference fits			Part no.
Customer:			Certificate no.
Order no. according to the specification of the customer:			Page: of
Manufacturer:			Work identification[1)]
Manufacturer's delivery no.:			1st line
Outer part	Customer	Manufacturer	2nd line
Drawing no.:			3rd line
Page and pos. no.:			Machine no.
Designation of workpieces:			
Material:			_____
Inner part	Customer	Manufacturer	
Drawing no.:			
Page and pos. no.:			
No. workpieces:			
Designation of workpieces: _____			Material: _____

1 Dimensional check						
Joint diameter	Nominal dimension	Tolerance class	Limit deviations	Actual measurements relative to left joint edge	centre of joint	right joint edge
Outer part (bore) Hole basic system						
Inner part (shaft)						

2 Surface texture (Drawing annotation according to DIN EN ISO 1302)			
Arithmetical roughness average *Ra* of joint surfaces	Nominal value	Actual value	Comment
Outer part (bore)			
Inner part (shaft)			

3 Cylindrical form tolerance (Drawing annotation according to DIN EN ISO 1101)			
Cylindrical form tolerance	Nominal value	Actual value	Comment
Outer part			
Inner part			

1) The hub has a marking on the front with the following information that is clearly and permanently visible:
- 1st line: Manufacturer's logo
- 2nd line: Manufacturer's no., drawing no., pos. no.
- 3rd line: Customer part no. (only for spare parts)

4 Joint temperature			
Joint temperature	°C Maximum value	°C Actual value	Comment
Outer part			
Inner part			

5 Checking for internal separation and surface cracks on the outer part after shrinkage

Test procedure:

Test result: *Free from cracks and internal separation that could affect usability.*

Comments:

Responsible for carrying out tests 1 to 4		Responsible for carrying out test 5	
(Place)	(Date)	(Place)	(Date)
(Signature) (Company stamp, testing organization)		(Signature) (Company stamp, testing organization)	

Annex D
(informative)

Explanatory notes

The elastic-plastic design of interference fits has not been used much in the past. However, as Kollmann shows in [2], it offers significant economic benefits, as the joint pressure can reach almost twice that of interference fits with purely elastic dimensions – especially with thick-walled hubs.

The calculation process according to Kollmann [4] and Kollmann and Önoz [3] that appeared in the July 1988 edition of this standard has been retained. It is based on the shear stress hypothesis (modified yield condition according to Tresca). Deviations from the distortion hypothesis frequently used in shaft design (yield condition according to von Mises) are smaller than the variations with the yield point R_e or $R_{p0,2}$. Thus, calculation programs based on the distortion hypothesis can also be used for the design of elastic-plastic interference fits.

The Kollmann und Önoz process [3] has the advantage that the effect of the centrifugal force can be recorded in rotating, interference fits subjected to elastic-plastic stresses [10]. Lundberg's theory [11], however, does not offer any solutions for this. Annex D includes several flow charts that aid the calculation process.

This standard includes an inspection certificate form in Annex C. This satisfies the documentation obligations regarding the safety aspects involved when using interference fits.

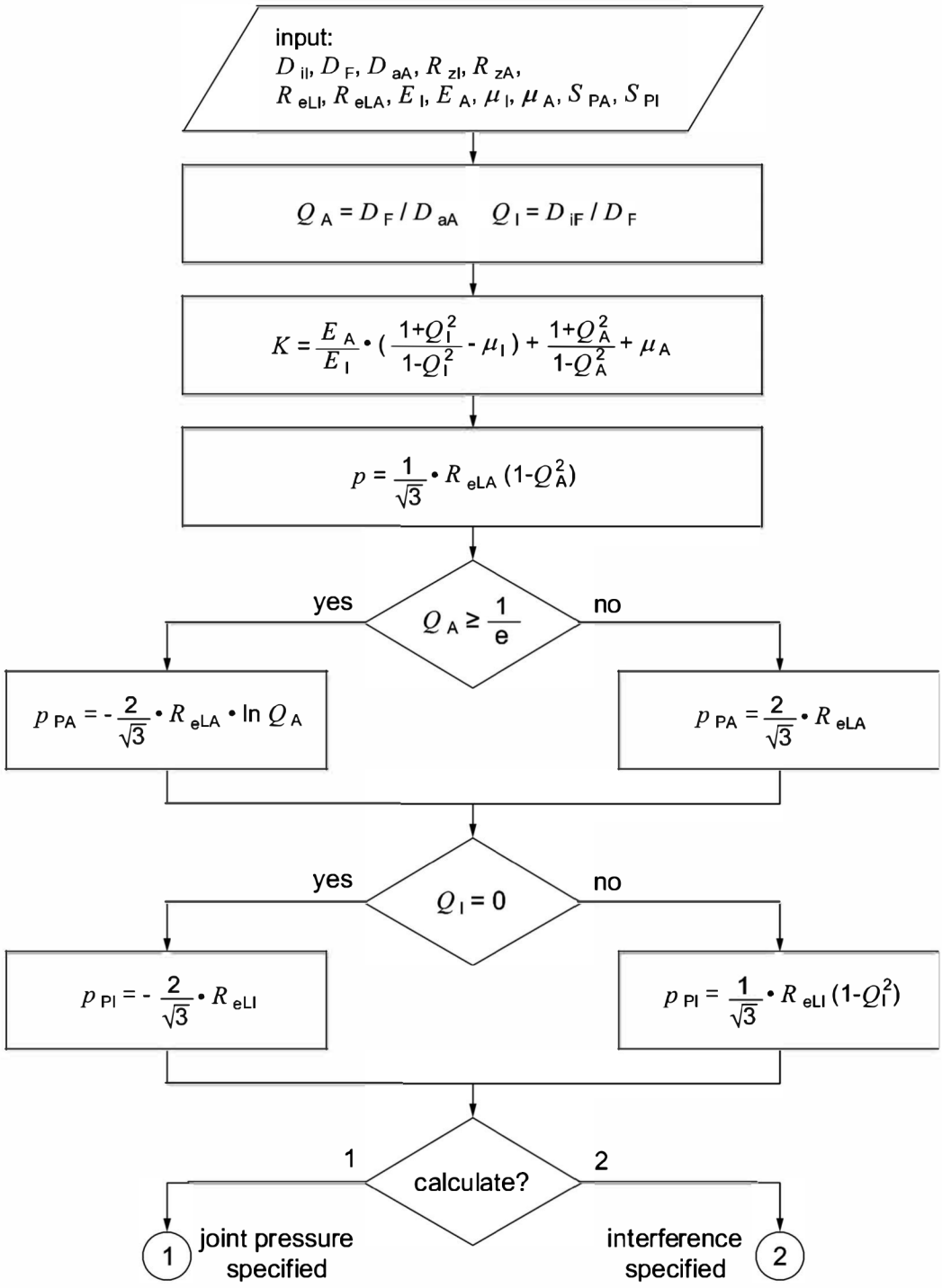


Figure D.1 — Flow chart for the calculation of elastic and elastic-plastic stresses in interference fits: Main program, Part 1 of 6

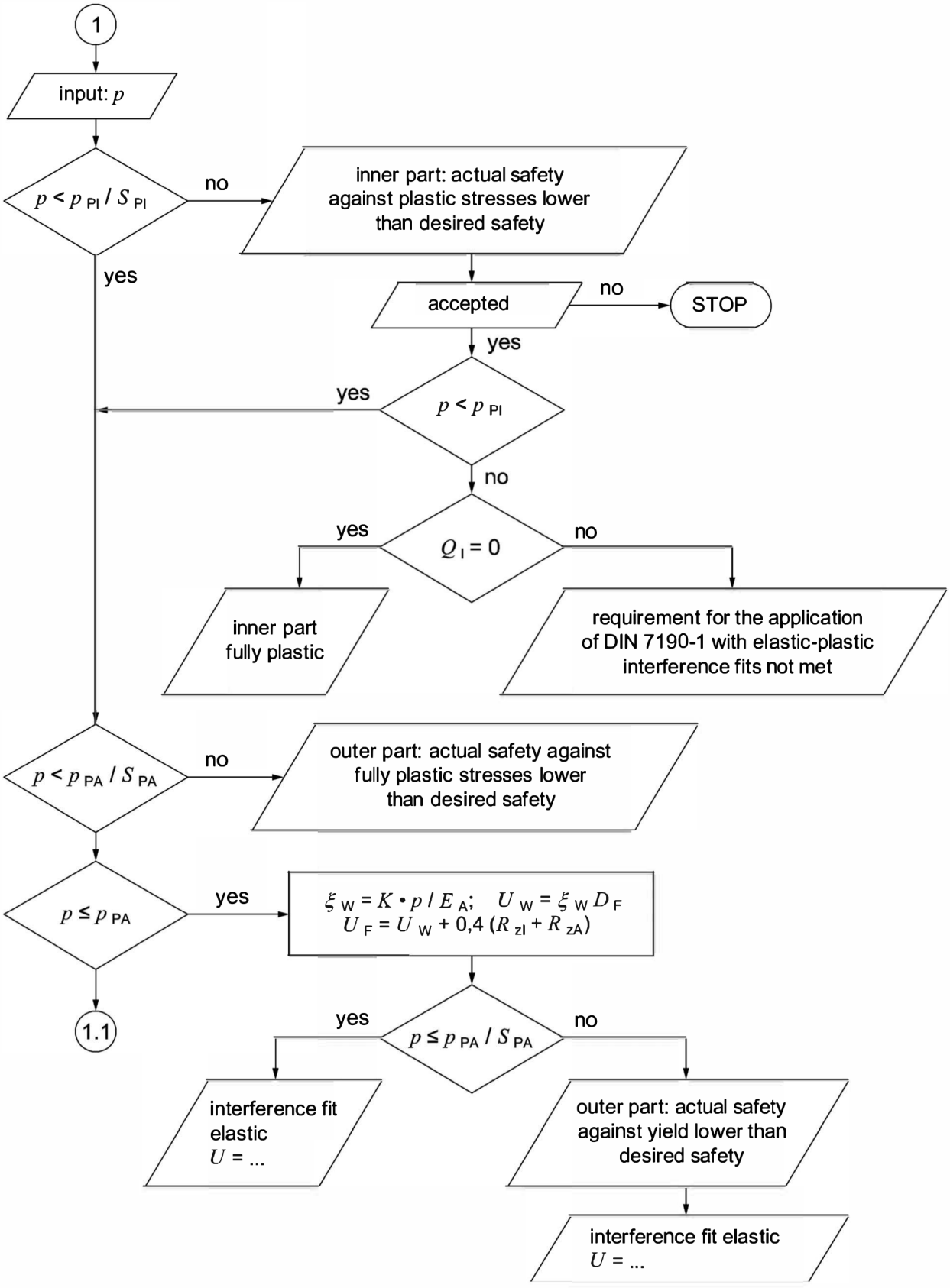


Figure D.2 — Flow chart for the calculation of elastic and elastic-plastic stresses in interference fits: Main program, Part 2 of 6

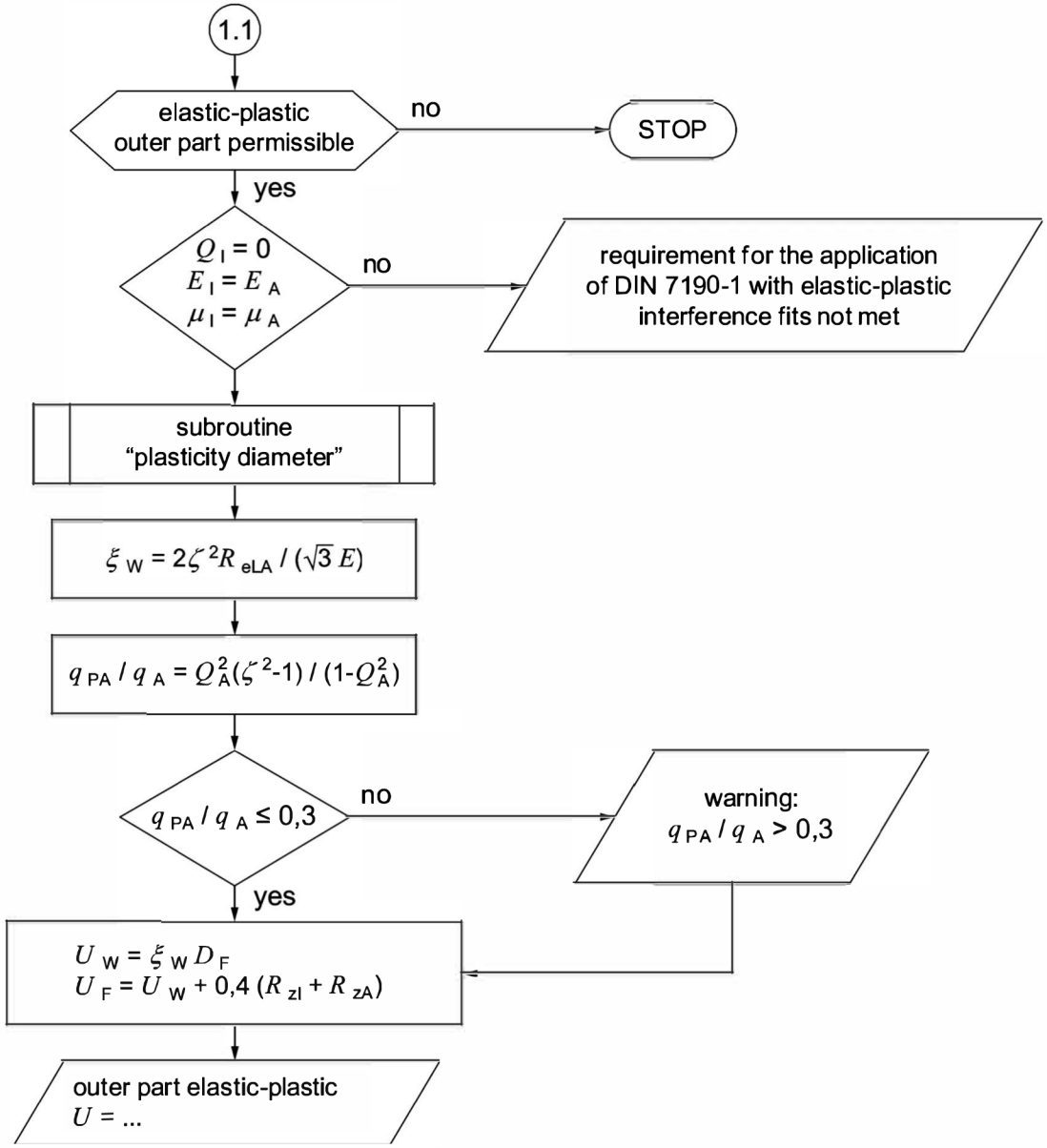


Figure D.3 — Flow chart for the calculation of elastic and elastic-plastic stresses in interference fits: Main program, Part 3 of 6

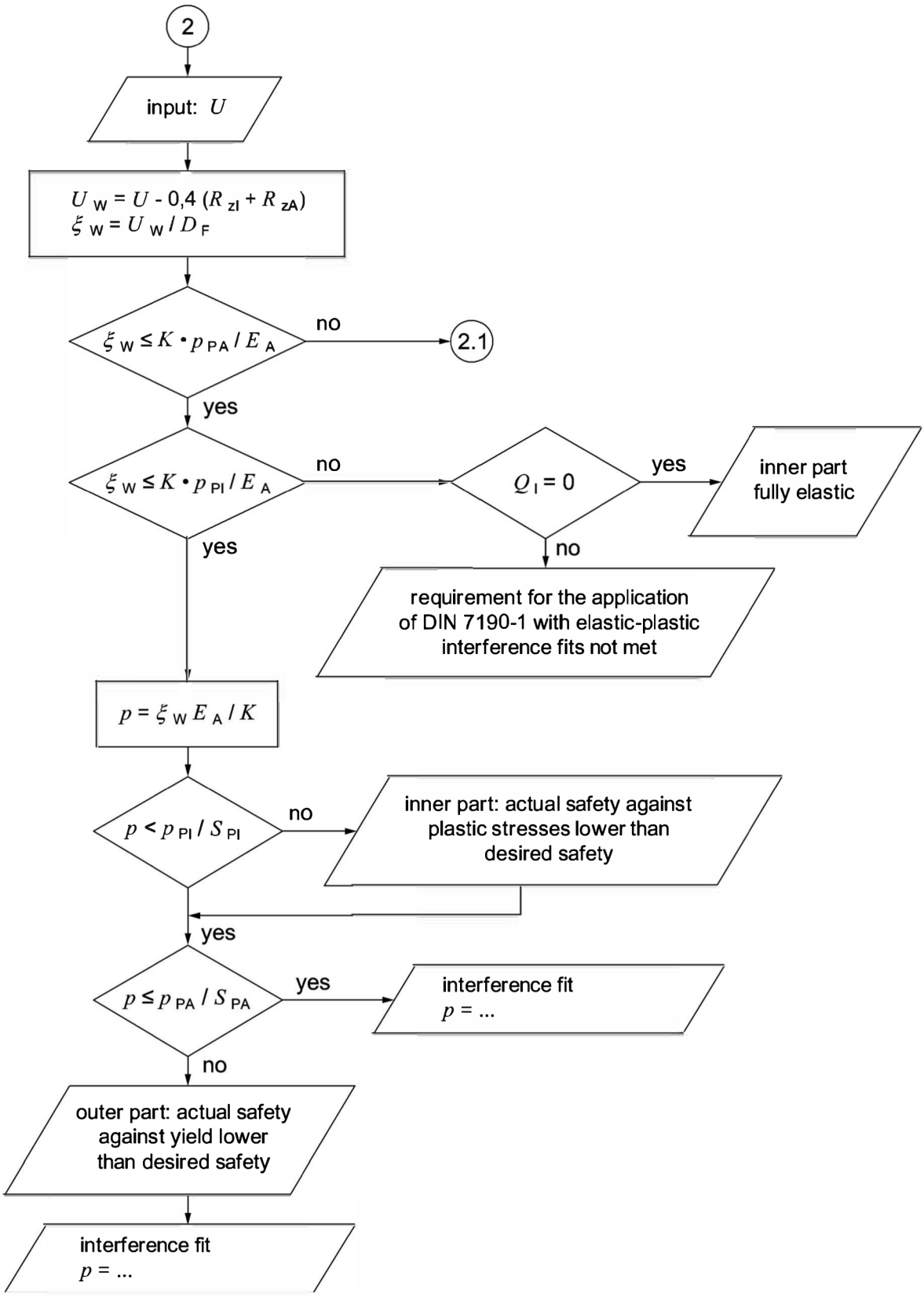


Figure D.4 — Flow chart for the calculation of elastic and elastic-plastic stresses in interference fits: Main program, Part 4 of 6

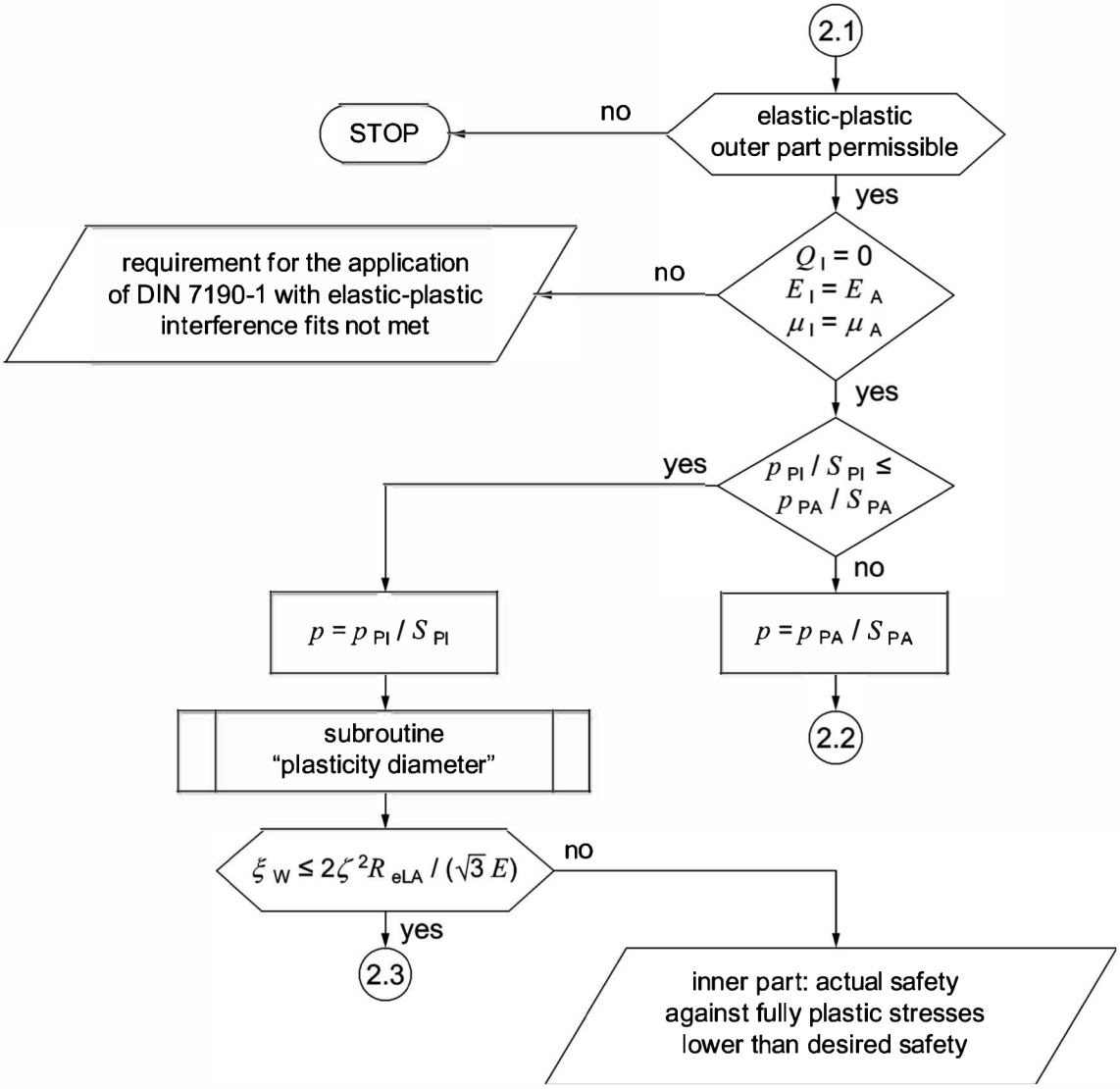


Figure D.5 — Flow chart for the calculation of elastic and elastic-plastic stresses in interference fits: Main program, Part 5 of 6

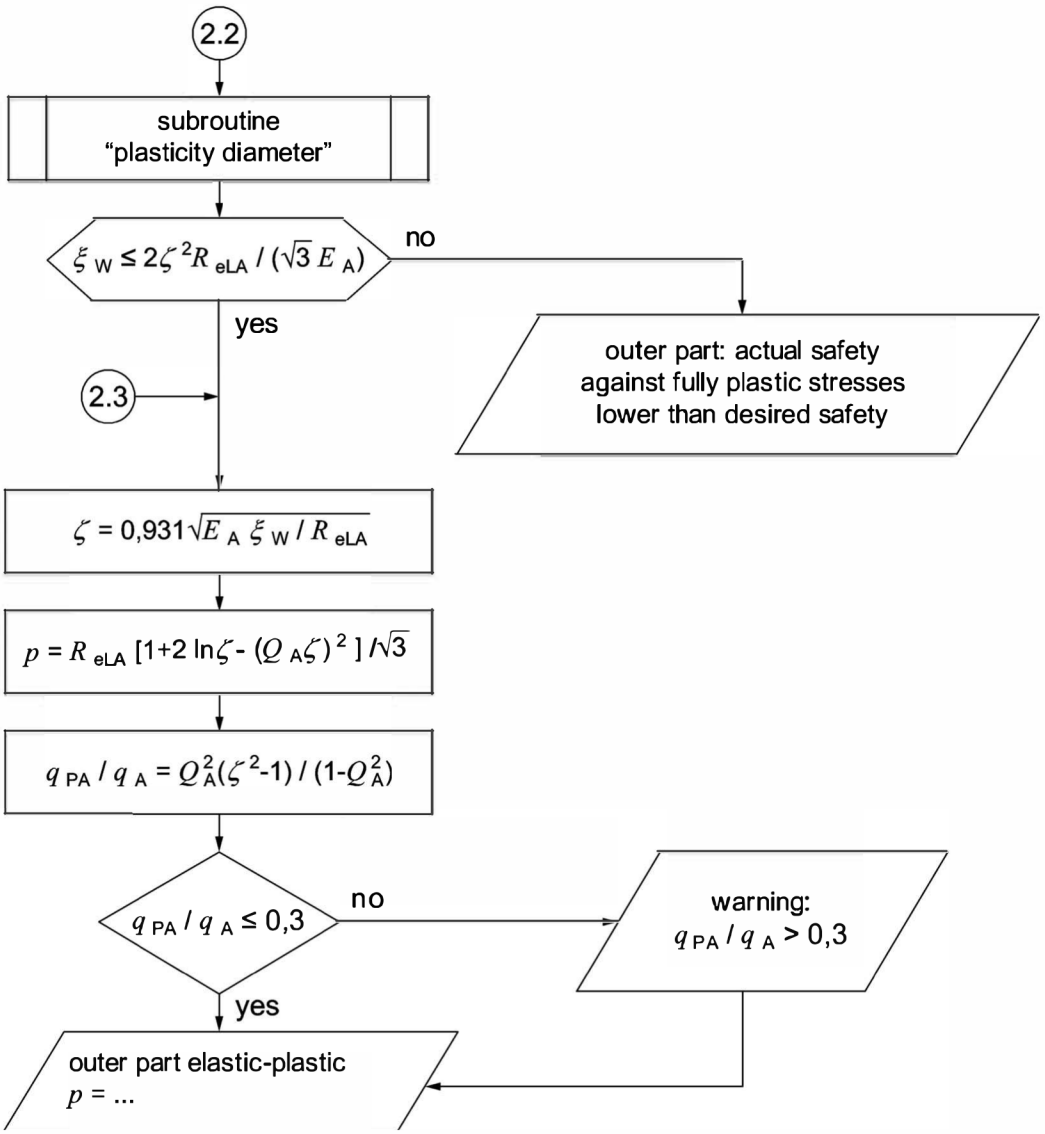


Figure D.6 — Flow chart for the calculation of elastic and elastic-plastic stresses in interference fits: Main program, Part 6 of 6

Parameters to be transferred from the main program: p, Q_A, R_{eLA}
Parameters to be transferred to the main program: ζ

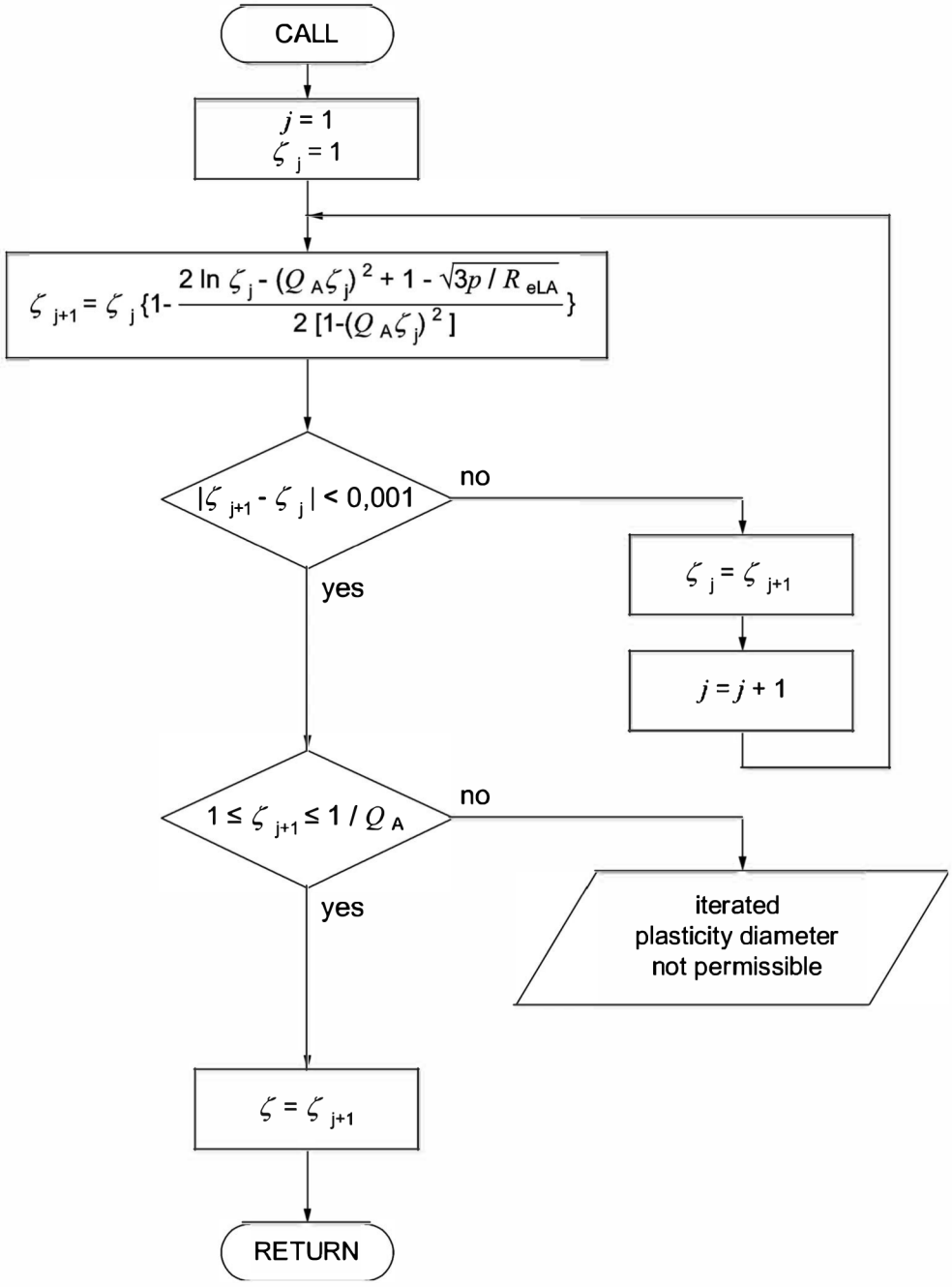


Figure D.7 — Flow chart for the calculation of elastic and elastic-plastic stresses in interference fits: Subroutine for the calculation of the plasticity diameter for specified joint pressure, p

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